

Field Effect Transistor - N-Channel, Enhancement Mode

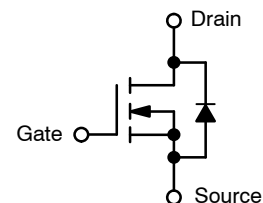
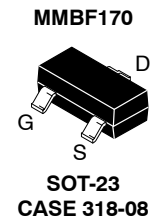
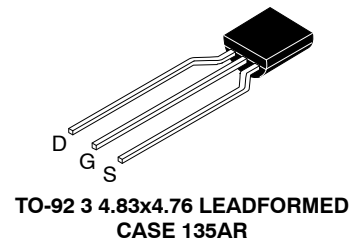
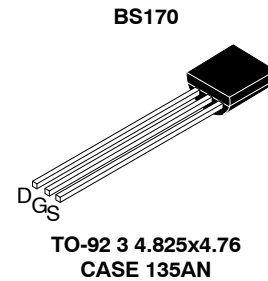
BS170, MMBF170

General Description

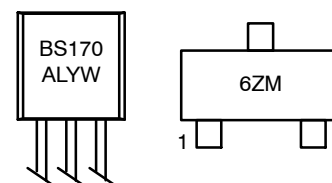
These N-Channel enhancement mode field effect transistors are produced using onsemi's proprietary, high cell density, DMOS technology. These products have been designed to minimize on-state resistance while provide rugged, reliable, and fast switching performance. They can be used in most applications requiring up to 500 mA DC. These products are particularly suited for low voltage, low current applications such as small servo motor control, power MOSFET gate drivers, and other switching applications.

Features

- High Density Cell Design for Low $R_{DS(ON)}$
- Voltage Controlled Small Signal Switch
- Rugged and Reliable
- High Saturation Current Capability
- These are Pb-Free Devices



MARKING DIAGRAM



BS170, 6Z = Device Code
A = Assembly Plant Code
L = Wafer Lot Number
YW = Assembly Start Week
M = Date Code

ORDERING INFORMATION

See detailed ordering and shipping information on page 8 of this data sheet.

NOTE: Some of the devices on this data sheet have been **DISCONTINUED**. Please refer to the table on page 8.

BS170, MMBF170

ABSOLUTE MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)

Symbol	Parameter		BS170	MMBF170	Unit
V _{DSS}	Drain-Source Voltage		60		V
V _{DGR}	Drain-Gate Voltage (R _{GS} ≤ 1 MΩ)		60		V
V _{GSS}	Gate-Source Voltage		±20		V
I _D	Drain Current	- Continuous	500	500	mA
		- Pulsed	1200	800	
T _J , T _{STG}	Operating and Storage Temperature Range		- 55 to 150		°C
T _L	Maximum Lead Temperature for Soldering Purposes, 1/16" from Case for 10 Seconds		300		°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)

Symbol	Parameter	BS170	MMBF170	Unit
P _D	Maximum Power Dissipation Derate above 25 °C	830 6.6	300 2.4	mW mW/°C
R _{θJA}	Thermal Resistance, Junction to Ambient	150	417	°C/W

ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)

Symbol	Parameter	Test Condition	Type	Min	Typ	Max	Unit
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OFF CHARACTERISTICS

BV _{DSS}	Drain-Source Breakdown Voltage	V _{GS} = 0 V, I _D = 100 μA	All	60	–	–	V
I _{DSS}	Zero Gate Voltage Drain Current	V _{DS} = 25 V, V _{GS} = 0 V	All	–	–	0.5	μA
I _{GSSF}	Gate - Body Leakage, Forward	V _{GS} = 15 V, V _{DS} = 0 V	All	–	–	10	nA

ON CHARACTERISTICS (Note 1)

V _{GS(th)}	Gate Threshold Voltage	V _{DS} = V _{GS} , I _D = 1 mA	All	0.8	2.1	3	V
R _{DS(on)}	Static Drain-Source On-Resistance	V _{GS} = 10 V, I _D = 200 mA	All	–	1.2	5	Ω
g _{FS}	Forward Transconductance	V _{DS} = 10 V, I _D = 200 mA	BS170	–	320	–	mS
		V _{DS} ≥ 2 V _{DS(on)} , I _D = 200 mA	MMBF170	–	320	–	

DYNAMIC CHARACTERISTICS

C _{iss}	Input Capacitance	V _{DS} = 10 V, V _{GS} = 0 V, f = 1.0 MHz	All	–	24	40	pF
C _{oss}	Output Capacitance		All	–	17	30	pF
C _{rss}	Reverse Transfer Capacitance		All	–	7	10	pF

SWITCHING CHARACTERISTICS (Note 1)

t _{on}	Turn-On Time	V _{DD} = 25 V, I _D = 200 mA, V _{GS} = 10 V, R _{GEN} = 25 Ω	BS170	–	–	10	ns
		V _{DD} = 25 V, I _D = 500 mA, V _{GS} = 10 V, R _{GEN} = 50 Ω	MMBF170	–	–	10	
t _{off}	Turn-Off Time	V _{DD} = 25 V, I _D = 200 mA, V _{GS} = 10 V, R _{GEN} = 25 Ω	BS170	–	–	10	ns
		V _{DD} = 25 V, I _D = 500 mA, V _{GS} = 10 V, R _{GEN} = 50 Ω	MMBF170	–	–	10	

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

1. Pulse Test: Pulse Width ≤ 300 μs, Duty Cycle ≤ 2.0%.



TYPICAL ELECTRICAL CHARACTERISTICS

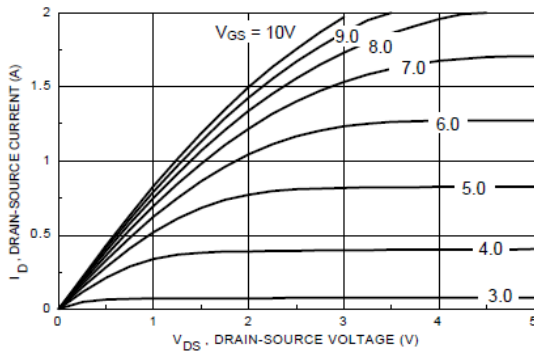


Figure 1. On-Region Characteristics

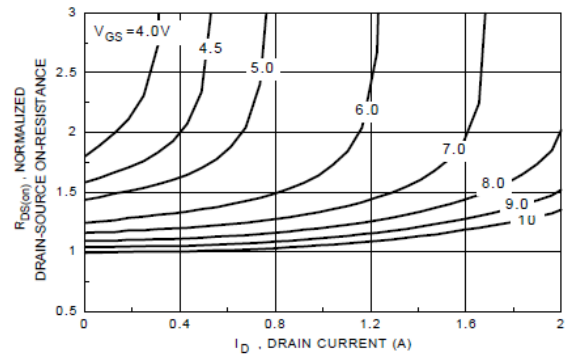


Figure 2. On-Resistance Variation with Gate Voltage and Drain Current

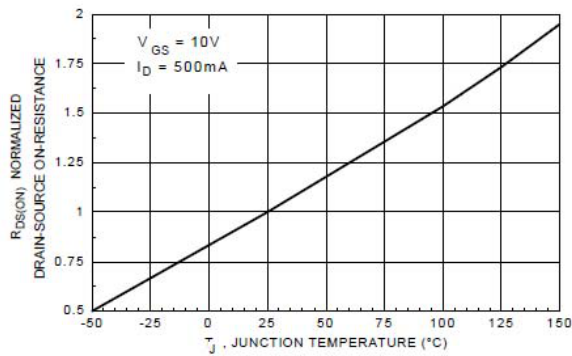


Figure 3. On-Resistance Variation with Temperature

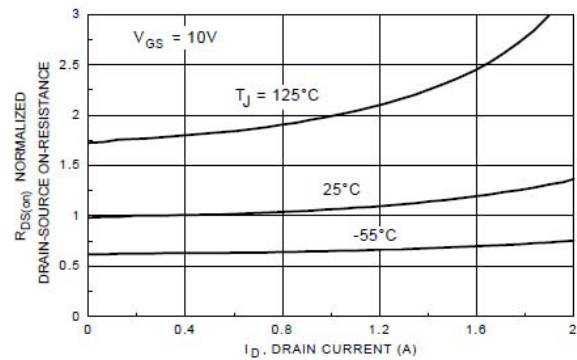


Figure 4. On-Resistance Variation with Drain Current and Temperature

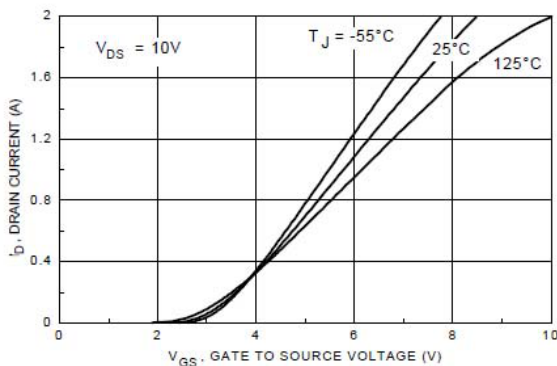


Figure 5. Transfer Characteristics

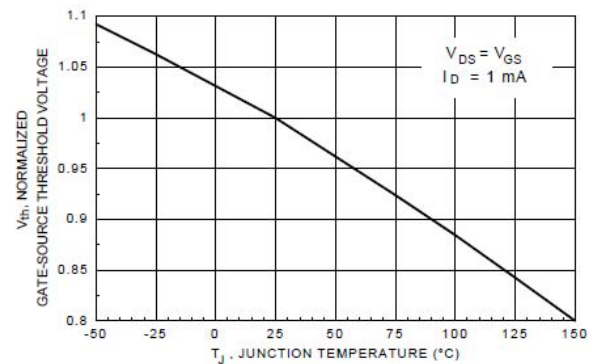


Figure 6. Gate Threshold Variation with Temperature

TYPICAL ELECTRICAL CHARACTERISTICS (CONTINUED)

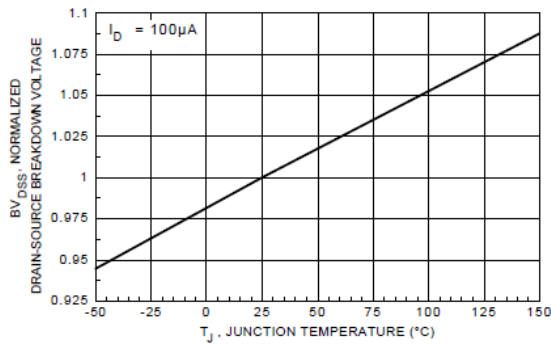


Figure 7. Breakdown Voltage Variation with Temperature

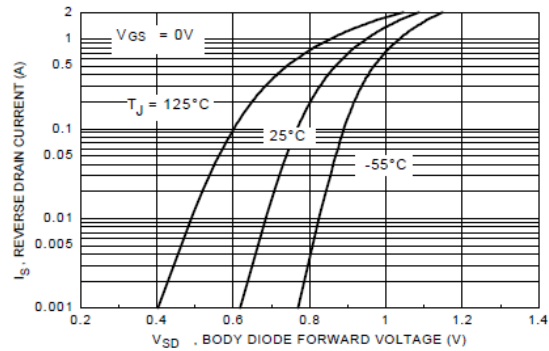


Figure 8. Body Diode Forward Voltage Variation with Current and Temperature

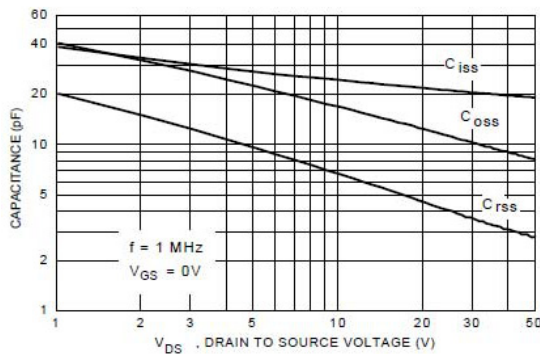


Figure 9. Capacitance Characteristics

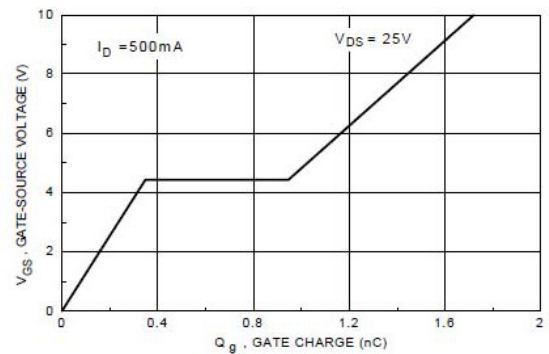


Figure 10. Gate Charge Characteristics

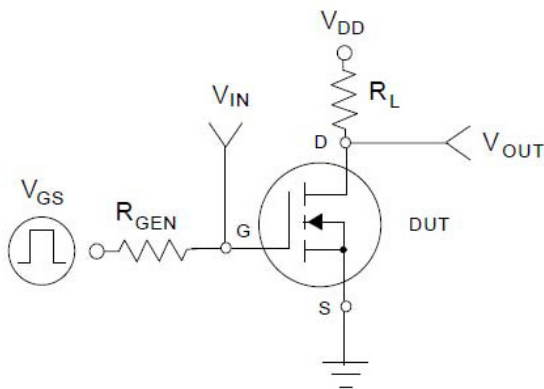


Figure 11. Switching Test Circuit

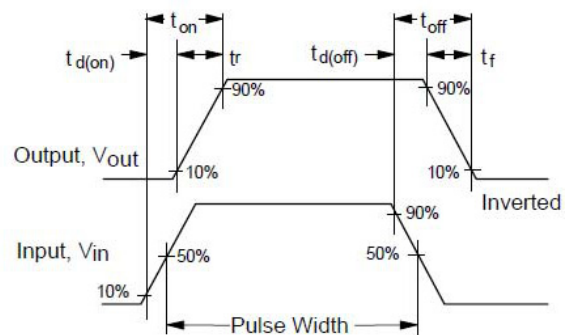


Figure 12. Switching Waveforms

BS170, MMBF170

TYPICAL ELECTRICAL CHARACTERISTICS (CONTINUED)

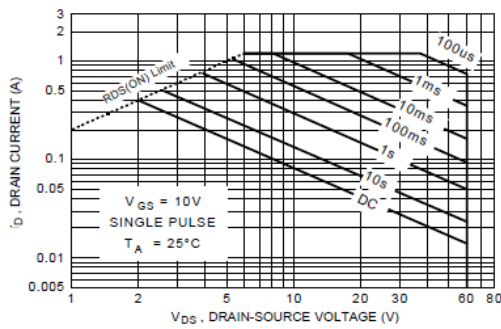


Figure 13. BS170 Maximum Safe Operating Area

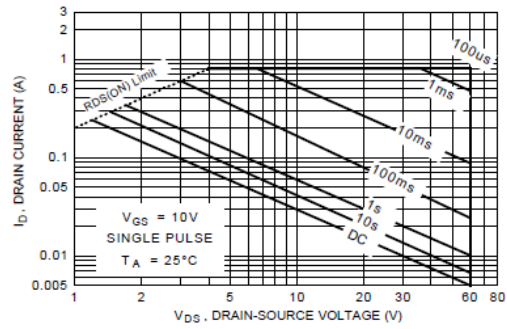


Figure 14. MMBF170 Maximum Safe Operating Area

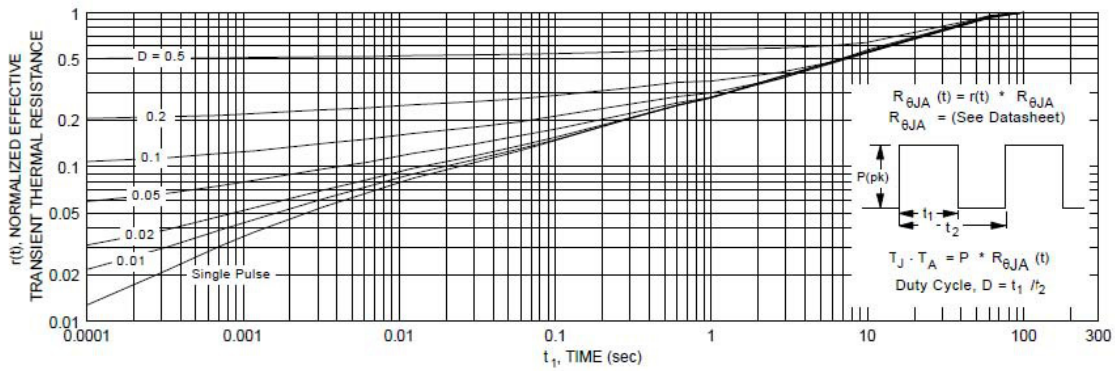


Figure 15. TO-92, BS170 Transient Thermal Response Curve

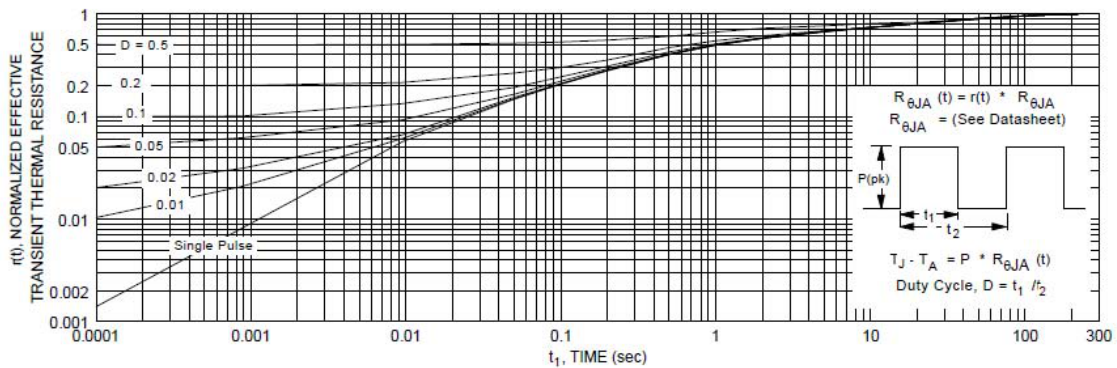


Figure 16. SOT-23, MMBF170 Transient Thermal Response Curve

BS170, MMBF170

ORDERING INFORMATION

Part Number	Package	Lead Frame	Pin Array	Shipping [†]
BS170	TO-92 (Pb-Free)	Straight	D G S	10,000 Units / Bulk
BS170-D26Z	TO-92 (Pb-Free)	Forming	D G S	2,000 / Tape & Reel
BS170-D27Z	TO-92 (Pb-Free)	Forming	D G S	2,000 / Tape & Reel
MMBF170	SOT-23 (Pb-Free)			3,000 / Tape & Reel

DISCONTINUED (Note 11)

BS170-D74Z	TO-92 (Pb-Free)	Forming	D G S	2,000 / Ammo
BS170-D75Z	TO-92 (Pb-Free)	Forming	D G S	2,000 / Ammo

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, [BRD8011/D](#).

2. **DISCONTINUED:** These devices are not available. Please contact your **onsemi** representative for information. The most current information on these devices may be available on www.onsemi.com.



BS170, MMBF170

REVISION HISTORY

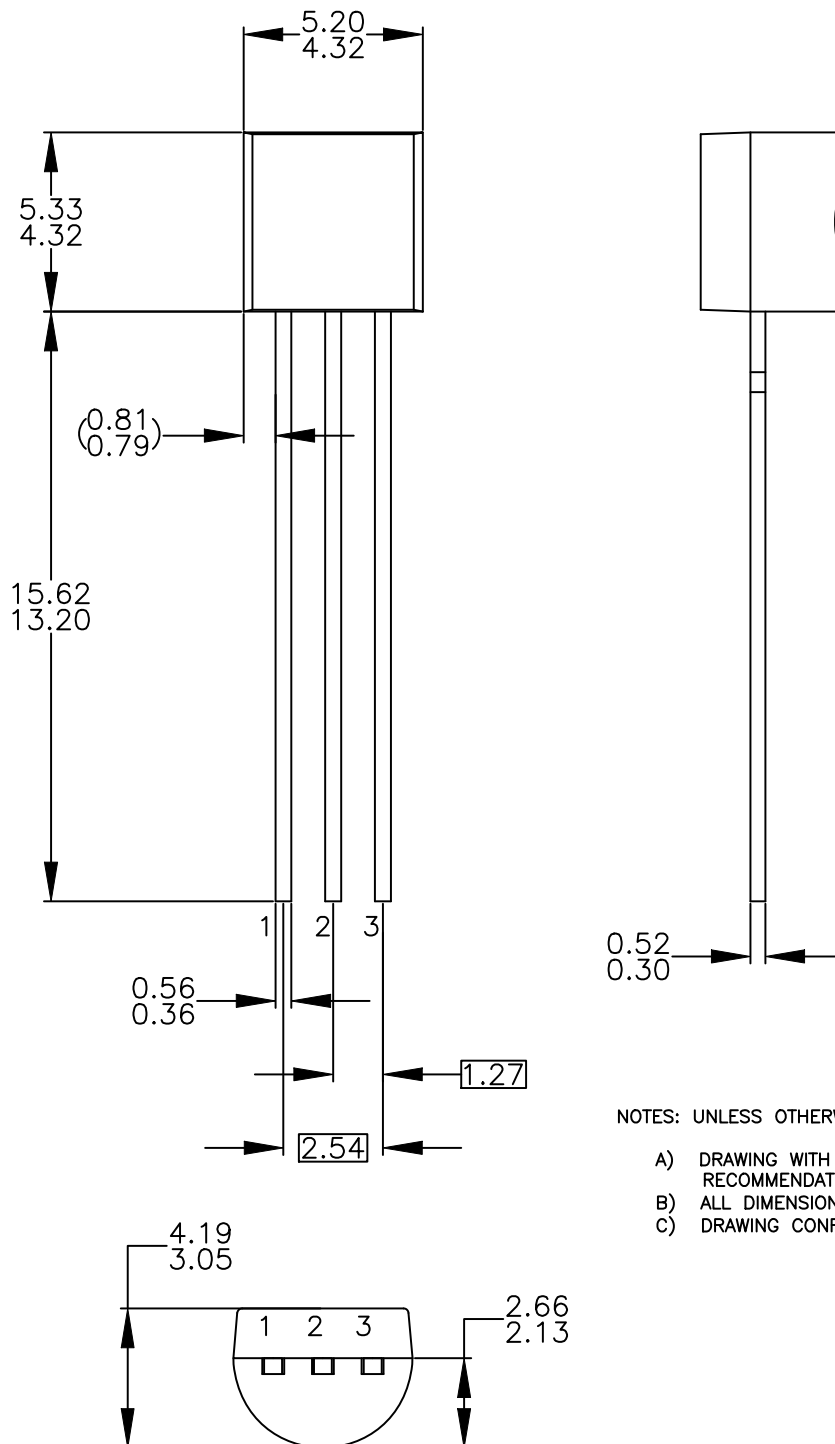
Revision	Description of Changes	Date
9	BS170-D75Z OPN Marked as Discontinued.	02/12/2026

This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.



TO-92 3 4.825x4.76
CASE 135AN
ISSUE O

DATE 31 JUL 2016



NOTES: UNLESS OTHERWISE SPECIFIED

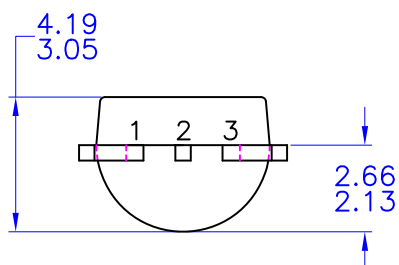
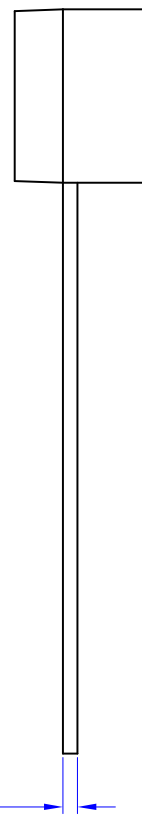
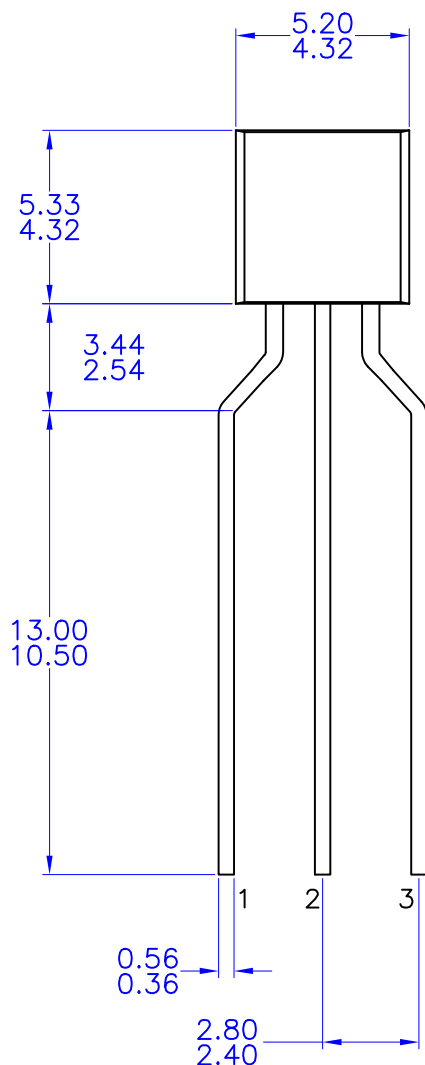
- A) DRAWING WITH REFERENCE TO JEDEC TO-92 RECOMMENDATIONS.
- B) ALL DIMENSIONS ARE IN MILLIMETERS.
- C) DRAWING CONFORMS TO ASME Y14.5M-2009.

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TO-92 3 4.83x4.76 LEADFORMED
CASE 135AR
ISSUE O

DATE 30 SEP 2016

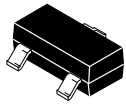


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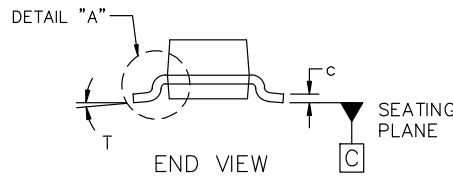
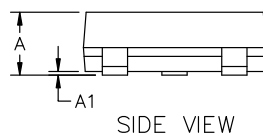
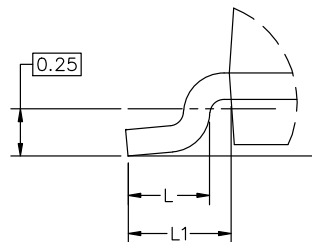
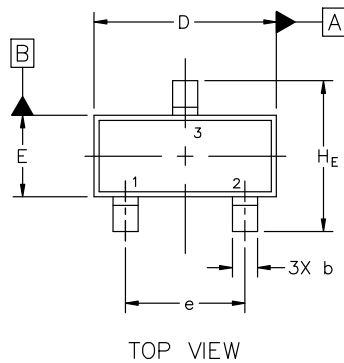
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SCALE 4:1

SOT-23 (TO-236) 2.90x1.30x1.00 1.90P
CASE 318
ISSUE AU

DATE 14 AUG 2024

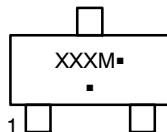


MILLIMETERS			
DIM	MIN	NOM	MAX
A	0.89	1.00	1.11
A1	0.01	0.06	0.10
b	0.37	0.44	0.50
c	0.08	0.14	0.20
D	2.80	2.90	3.04
E	1.20	1.30	1.40
e	1.78	1.90	2.04
L	0.30	0.43	0.55
L1	0.35	0.54	0.69
HE	2.10	2.40	2.64
T	0°	---	10°

NOTES:

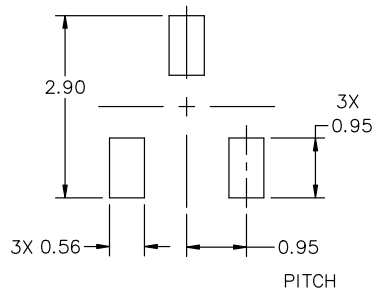
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSIONS: MILLIMETERS.
3. MAXIMUM LEAD THICKNESS INCLUDES LEAD FINISH. MINIMUM LEAD THICKNESS IS THE MINIMUM THICKNESS OF THE BASE MATERIAL.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS.

GENERIC MARKING DIAGRAM*



XXX = Specific Device Code
M = Date Code
▪ = Pb-Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb-Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.



* For additional information on our Pb-Free strategy and soldering details, please download the onsemi Soldering and Mounting Techniques Reference Manual, SOLDERM/D.

STYLES ON PAGE 2

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CASE 318
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DATE 14 AUG 2024

STYLE 1 THRU 5: CANCELLED	STYLE 6: PIN 1. BASE 2. EMITTER 3. COLLECTOR	STYLE 7: PIN 1. EMITTER 2. BASE 3. COLLECTOR	STYLE 8: PIN 1. ANODE 2. NO CONNECTION 3. CATHODE		
STYLE 9: PIN 1. ANODE 2. ANODE 3. CATHODE	STYLE 10: PIN 1. DRAIN 2. SOURCE 3. GATE	STYLE 11: PIN 1. ANODE 2. CATHODE 3. CATHODE-ANODE	STYLE 12: PIN 1. CATHODE 2. CATHODE 3. ANODE	STYLE 13: PIN 1. SOURCE 2. DRAIN 3. GATE	STYLE 14: PIN 1. CATHODE 2. GATE 3. ANODE
STYLE 15: PIN 1. GATE 2. CATHODE 3. ANODE	STYLE 16: PIN 1. ANODE 2. CATHODE 3. CATHODE	STYLE 17: PIN 1. NO CONNECTION 2. ANODE 3. CATHODE	STYLE 18: PIN 1. NO CONNECTION 2. CATHODE 3. ANODE	STYLE 19: PIN 1. CATHODE 2. ANODE 3. CATHODE-ANODE	STYLE 20: PIN 1. CATHODE 2. ANODE 3. GATE
STYLE 21: PIN 1. GATE 2. SOURCE 3. DRAIN	STYLE 22: PIN 1. RETURN 2. OUTPUT 3. INPUT	STYLE 23: PIN 1. ANODE 2. ANODE 3. CATHODE	STYLE 24: PIN 1. GATE 2. DRAIN 3. SOURCE	STYLE 25: PIN 1. ANODE 2. CATHODE 3. GATE	STYLE 26: PIN 1. CATHODE 2. ANODE 3. NO CONNECTION
STYLE 27: PIN 1. CATHODE 2. CATHODE 3. CATHODE	STYLE 28: PIN 1. ANODE 2. ANODE 3. ANODE				

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